

Cory Pearl

(571) 447-0852 | cp3349@drexel.edu | github.com/CoryPearl | linkedin.com/in/cory-pearl

EDUCATION

Drexel University

Bachelor of Science in Computer Science

GPA: 3.22

Philadelphia, PA

Anticipated: June 2030

SKILLS

Software: AWS, EC2, Vercel, Figma, Fusion 360, Arduino, MySQL, node.js, Linux, Excel, Git, Github, apache, nginx, Claude, cloudflare R2, Codex, Linux Command line. FreeRTOS, ESP-IDF, KiKad, EasyEDA, FreeCAD

Languages: Python, Javascript, HTML/CSS, C, Swift, AArch64 & x86 Assembly, SQL, Bash, System verilog

Hardware: Arduino/Arduino Nano, ESP32, Raspberry Pi, DigiSpark, ATtiny85, Soldering

PROJECTS

OpenCalc – Open-Source Graphing Calculator (ESP32-S3, ESP-IDF, KiKad)

May to July 2026

- Developed a custom graphing calculator firmware and custom PCB in KiKad, featuring graphing, statistics, matrix math, USB storage, and Python-style scripting support
- Implemented multitasking firmware using FreeRTOS and integrated LCD graphics, touch input, keypad scanning, and embedded applications on an ESP32-S3 platform and designed a custom PCB for it

Flick It Philly - AI-Powered 311 Reporting App React Native, Node.js, Gemini API

March 2026

- Built a mobile app with a team that streamlines Philadelphia 311 issue reporting by using AI vision to automatically classify and describe infrastructure problems from a user-submitted photo
- Architected a client-server pipeline integrating the device camera, a Node.js backend, and the Gemini API

CodeSync - Collaborative Browser-Based Coding Platform Node.js, WebSockets, JavaScript

June 2024

- Built a full stack, real-time multi-user code editor with in-browser execution, synchronizing editor state across clients using a WebSocket event system
- Designed concurrent session handling to prevent state conflicts between simultaneous users

EXPERIENCE

Regional President

June 2024 to June 2025

B'nai Brith Youth Organization

Washington Metropolitan Area

- Led collaboration among regional, national, and international youth leaders to coordinate programs and initiatives across a multi-chapter region
- Worked closely with professional staff to oversee regional operations, strategic planning, and program execution, serving as the primary liaison between youth and adult leadership
- Managed and guided a regional board, directing marketing, recruitment efforts, and large-scale programming

Camp Counselor/Lifeguard

June 2025 to August 2025

Camp Airy

Thurmont, MD

- Supervised campers 24/7 in a residential setting, fostering a culture of teamwork, responsibility, and inclusion
- Led structured activities and games that encouraged creativity and engagement across diverse age groups
- Ensured safety and responded to emergencies with vigilance, empathy, and quick decision-making

Lifeguard

October 2023 to June 2025

Arlington Public Schools

Arlington, VA

- Monitored aquatic facilities to maintain swimmer safety and enforce pool regulations across multiple school sites
- Responded promptly to emergencies, administering first aid and coordinating with staff as needed
- Maintained organized, hazard-free pool environments and proactively communicated safety guidelines

Teaching Assistant

June 2021 to August 2023

Congregation Etz Hayim

Arlington, VA

- Assisted lead teachers with instruction and activity preparation, developing supplementary materials tailored to diverse learning styles
- Provided individualized and small-group support to reinforce lesson content and build student confidence
- Helped manage classroom dynamics, ensuring all students felt safe, respected, and included

RELEVANT COURSEWORK

Comp & Info Design I & II & III, Computer Programming I & II, Multivariate Calculus, Calculus II & III, Physics I & II